

Interview Questions

1) Explain the Components in df -h ?

```
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        50G   20G   28G   42% /
tmpfs            2.0G     0  2.0G    0% /dev/shm
/dev/sdb1       100G   60G   35G   64% /mnt/data
```

2) Explain the Components in free -m ?

```
$ free -m
              total        used        free      shared  buff/cache   available
Mem:           7982         5120         1024          200         1838         2536
Swap:          2048           0         2048
```

3) How to see the running process of particular site ? Example (Chrome)

A) ps aux | grep Chrome aux – all user extended

4) Explain components in Top Command?

A)

```
top - 18:26:40 up 20 days, 3:48, 1 user, load average: 1.06, 1.18, 1.56
Tasks: 337 total, 2 running, 335 sleeping, 0 stopped, 0 zombie
%Cpu(s): 9.3 us, 3.0 sy, 0.0 ni, 85.7 id, 2.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7699.5 total, 137.4 free, 6411.4 used, 1150.7 buff/cache
MiB Swap: 5897.7 total, 225.7 free, 5672.0 used. 443.6 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 3325 adminis+  20   0 5903612 167292 24648 R   16.7   2.1 340:22.43 gnome-shell
 3193 adminis+  20   0 839340 28200 5180 S    8.7   0.4 259:31.95 Xorg
3028484 adminis+  20   0 836836 21760 10040 S    4.0   0.3 1:57.12 gnome-terminal-
2932480 adminis+  20   0 33.5g 476384 140408 S    3.3   6.0 379:23.73 chrome
3180832 adminis+  20   0 1132.0g 494504 90716 S    3.3   6.3 15:30.80 chrome
 4188 adminis+  20   0 1123.0g 42940 27640 S    3.0   0.5 18:50.60 cliq
 4380 adminis+  20   0 1134.9g 124584 38336 S    2.7   1.6 465:22.38 cliq
2932524 adminis+  20   0 33.0g 181400 81228 S    2.3   2.3 443:20.27 chrome
 4315 adminis+  20   0 32.5g 31324 13992 S    1.7   0.4 363:25.04 cliq
 197 root      -SI   0      0      0      0 S    1.0   0.0 13:25.20 /usr/bin/ssh-agent
```

5) What is Swap Usage ?

A) Swap usage is temporary extension of RAM, when RAM is full we used swap usage.

6) Explain 5 start in scheduling Jobs ?

A)



7) Difference between Application Load balancer and Network load balancer ?

A)

Feature	ALB (Application Load Balancer)	NLB (Network Load Balancer) 
Layer	Layer 7 (Application Layer – HTTP/HTTPS)	Layer 4 (Transport Layer – TCP/UDP)
Use Case	Web applications, routing URLs or hostnames	High-performance, low-latency apps
Smart Routing	✔ Yes (based on path, hostname, headers, etc)	✘ No (just forwards traffic)
Protocol Support	HTTP, HTTPS, WebSocket	TCP, TLS, UDP
Speed & Performance	Good, slightly slower due to smart routing	Very fast, optimized for high throughput

8) Which Load balancer is used access another Load balancer ?

A) By Using Network Load balancer we can access another Load balancer.

9) What is DNS ?

A) DNS (Domain Name System) is service that translates Domain name into IP address.

10) Difference between Security group and NACL ?

A)

Feature	Security Group (SG)	Network ACL (NACL)
Type	Instance-level firewall	Subnet-level firewall
State	Stateful – return traffic is automatically allowed	Stateless – return traffic must be allowed manually
Rules	Only allow rules	Can allow or deny traffic
Applies To	Attached to EC2 instances	Applied to subnets

11) Where we can store Terraform state file ?

A) Terraform state files can be stored locally by default or remotely using backends like AWS S3, Terraform Cloud, or Azure Blob. For team collaboration, it's best to use **S3 with DynamoDB** for state locking to avoid conflicts.

```
terraform {  
  backend "s3" {  
    bucket     = "my-terraform-state-bucket"    # S3 bucket name  
    key        = "env/dev/terraform.tfstate"    # Path inside the bucket  
    region     = "us-east-1"                   # AWS region  
    dynamodb_table = "terraform-locks"         # DynamoDB table for locking  
    encrypt    = true                           # Encrypt the state file at rest  
  }  
}
```

12) Difference between Deployment , Stateful , Demonset ?

A)

Resource	Purpose / Key Difference
Deployment	For stateless apps (e.g. web apps). Automatically scales and replaces pods.
StatefulSet	For stateful apps (e.g. databases). Maintains pod order, name, and storage.
DaemonSet	Runs one pod on each node (e.g. monitoring, logging agents).

13) What is Stateful and Stateless ?

- A) **Stateless applications** don't store any user data between requests, making them easy to scale. **Stateful applications** keep track of user data or session information, requiring persistent storage and careful scaling.

14) what is git life-cycle ?

A) Git Life-cycle in 4 Simple Steps:

1. **Working Directory:** You create or edit files locally on your computer.
2. **Staging Area (Index):** You add changes to a staging area using `git add` to prepare them for commit.
3. **Local Repository:** You save your changes permanently with `git commit`.
4. **Remote Repository:** You push your commits to a shared server (like GitHub) using `git push`.

15) Common work between Development team and operation team ?

- A) The development team writes and updates the application code, while the operations team manages deployment, infrastructure, and system reliability. Both teams collaborate to ensure smooth, continuous delivery and quick issue resolution. This collaboration is the core of DevOps culture.

16) How to install nginx in docker ?

- A) `docker run --name container name -d -p 80:80 nginx:latest`

17) How do you install docker in your host ?

- A) `apt install docker.io -y`

18) How do you start docker in host server ?

- A) `systemctl start docker`

19) What are Plugins used in Jenkins ?

A) Git Checkout , ECR, Docker, Docker pipeline, Email Extension, AWS Credentials, Nexus Artifactory, SonarQube Scanner, EKS .

20) How do you Create Jenkins using Docker ?

A) Docker run --name container name -d -p 8080:8080 Jenkins : latest

21) Explain Docker file commands ?

A)

Command	Purpose (Simple Explanation)
FROM	Sets the base image (e.g., <code>ubuntu</code> , <code>node</code> , <code>python</code>).
RUN	Runs commands in the image during build (e.g., install tools).
COPY	Copies files/folders from your system into the image.
ADD	Like <code>COPY</code> , but can also handle URLs and extract tar files.
CMD	Sets the default command to run when the container starts.
ENTRYPOINT	Similar to <code>CMD</code> , but more fixed — always runs this command.
WORKDIR	Sets the working directory inside the container.
EXPOSE	Tells Docker which port the app listens on.
ENV	Sets environment variables inside the container.

22) What is VPC and Components ?

A) VPC is our own private network inside cloud like AWS.

Components : Subnets, Route table, Internet-gateway, NAT gate-way, Security groups, NACL

23) what is Route 53 ?

A) **Amazon Route 53** is a **DNS (Domain Name System) service** from AWS. It helps you connect domain names to your servers, websites, or AWS resources. It also supports traffic routing, health checks, and domain registration.

24) I created one EC2 instance and I am trying to connect internet but I am getting error . How to resolve this and what you need to basic steps ?

A) If an EC2 instance can't access the internet, I check the subnet ,route table, ensure an Internet

Gateway is attached, assign a public IP, and verify that the security group and NACL allow outbound traffic.

25) I created one EC2 instance and I am trying to connect git bash ssh but I am getting error . How to resolve this and what you need to basic steps ?

A) If I can't SSH into my EC2 instance using Git Bash, I first check the **key pair (.pem file)** permissions, make sure I'm using the **correct public IP**, and verify that **port 22 is open** in the **security group**. I also ensure the instance is in a **public subnet** with an **Internet Gateway**.

26) How to attach EC2 to S3 ?

A) To allow EC2 to access S3, I create an **IAM role** with **AmazonS3FullAccess** or custom S3 policy, then attach that role to the EC2 instance. This way, the instance can access S3 securely without needing access keys.

27) When I am connecting to S3 getting error how to resolve and what we need to do basic steps ?

A) If EC2 can't access S3, I check if the IAM role is attached with correct S3 permissions, ensure the bucket policy allows it, and verify I'm using the correct region and CLI setup.

28) I wrote Terraform Script uploaded into GitHub where the credentials will stored ?

A) 1) In local I am storing in .aws credential file
2) GitHub secretes in Repository settings.
3) stores secretes in Terraform workspace

29) What is Terraform state file ?

A) **Terraform state file** (`terraform.tfstate`) is a file that keeps track of the **real-world infrastructure** Terraform manages. It stores the current state of your resources (like EC2, S3, etc.) so Terraform knows what exists and what needs to be changed during future runs.

30) Write one sample docker file and Explain ?

A) # Use the official NGINX image as the base
FROM nginx:latest

Copy custom HTML file into the NGINX default directory
COPY index.html /usr/share/nginx/html/index.html

Expose port 80 to allow web traffic
EXPOSE 80

31) How to Create Docker image ?

A) First we need to Create docker file .

docker build -t image name .

32) How to create Docker container ?

A) docker run --name container name -d -p port no:port no image name:tag

33) How to delete unused all docker containers at time ?

A) docker container prune -f

34) How to Terraform attached to AWS?

- A) 1. **Install Terraform**
2. **Set Up AWS CLI**
3. Create Access key in IAM
4. Create a Terraform Configuration File

35) What is Difference Virtualization and Containerization ?

A)

Feature	Virtualization	Containerization
Definition	Creates multiple virtual machines (VMs) on a single physical machine.	Runs multiple lightweight containers on a single host using the OS kernel.
Technology Used	Hypervisor (e.g., VMware, VirtualBox, KVM)	Container Engine (e.g., Docker, containerd)
OS Sharing	Each VM has its own OS.	Containers share the host OS kernel.
Resource Usage	Heavyweight – more CPU, RAM, and storage.	Lightweight – uses fewer resources.
Boot Time	Slower – takes minutes.	Faster – starts in seconds.
Isolation	Strong isolation (at hardware level).	Lighter isolation (at process level).

36) I got CPU Utilization is 100% for my instance,how resolve this issue ?

- A) 1. Change the instance type (Vertical Scaling immediate fix)
2. Use horizontal Auto-scaling (Long Term fix)
3. Check the running process of particular instance and Kill the running process which is having unwanted storage and restart the Server.

37) What is upstream and downstream in Jenkins?

A) In Jenkins, an *upstream* job is one that triggers another job, and a *downstream* job is the one that gets triggered.

For example : if Job A builds the code and then triggers Job B to deploy it, Job A is upstream and Job B is downstream. This helps in automating CI/CD pipelines in a proper sequence.

38) Difference between DevOps and DevsecOps ?

A)

Aspect	DevOps	DevSecOps
Focus	Development + Operations	Development + Security + Operations
Goal	Faster software delivery with automation	Fast delivery with built-in security
Security	Added after the development process	Security is integrated from the start
Who handles it	Developers and Operations teams	Developers, Security, and Operations teams
Main Tools	Jenkins, Docker, Kubernetes, etc.	Includes DevOps tools plus security tools (e.g., SonarQube, Snyk, Aqua)

39) I have 10 stages in my pipeline. The error occurred in stage 6. I don't want to rerun the pipeline from the beginning — I want to continue from stage 6 ?

A) 1. using Checkpoint Plugin but it works in scripted pipeline.

2. using the **when** condition and a **build parameter** to control which stage to start from.

```
pipeline {
    agent any

    parameters {
        string(name: 'START_FROM_STAGE', default Value: '6', description: 'Stage number to start from
        (e.g., 6 to skip first 5 stages)')
    }

    stages {
```



```

stage('Stage 1') {
    when { expression { params.START_FROM_STAGE.toInteger() <= 1 } }
    steps {
        echo "Running Stage 1"
    }
}
stage('Stage 2') {
    when { expression { params.START_FROM_STAGE.toInteger() <= 2 } }
    steps {
        echo "Running Stage 2"
    }
}
stage('Stage 3') {
    when { expression { params.START_FROM_STAGE.toInteger() <= 3 } }
    steps {
        echo "Running Stage 3"
    }
}
stage('Stage 4') {
    when { expression { params.START_FROM_STAGE.toInteger() <= 4 } }
    steps {
        echo "Running Stage 4"
    }
}
stage('Stage 5') {
    when { expression { params.START_FROM_STAGE.toInteger() <= 5 } }
    steps {
        echo "Running Stage 5"
    }
}
stage('Stage 6') {
    when { expression { params.START_FROM_STAGE.toInteger() <= 6 } }
    steps {
        echo "Running Stage 6"
    }
}
// Add more stages similarly...
}
}

```

40) What is Docker Architecture ?

A) Docker architecture has **3 main components**:

1. Docker Client :

- The **user interface** for Docker.
- You type commands here like `docker build`, `docker run`, `docker push`.

- The client talks to the Docker daemon to execute your commands.

2. Docker Daemon :

- The **engine** running on your machine or server.
- Listens for Docker API requests from the client.
- Manages Docker objects like images, containers, networks, and volumes.
- It builds, runs, and manages containers.

3. Docker Registry :

- A place to store Docker images.
- Can be **public** (like Docker Hub) or **private**.
- You **pull** images from here to your local system and **push** your custom images here.

41) What is pod and Cluster in Kubernetes ?

A) Pod : A Pod is the smallest deployable unit in Kubernetes, which encapsulates one or more containers, shared storage, and networking .

Cluster : A **Kubernetes Cluster** is a group of nodes that work together to run containerized applications. It manages and runs your apps reliably by distributing work across these machines.

42) Explain Kubernetes architecture ?

A) 1. **Control Plane (Master Node)**

- This is the brain of the Kubernetes cluster.
- It manages the cluster, decides what runs where, and handles the overall state.
- Key components inside the Control Plane:
 - **API Server:** The front door — it receives commands (like `kubectl` commands) and processes them.
 - **etcd:** The database that stores the cluster's state and configuration.
 - **Scheduler:** Decides which worker node will run a new Pod.
 - **Controller Manager:** Keeps the cluster running by handling things like replication and node failures.

2. Worker Nodes

- These are the machines where your applications actually run.

- Each worker node runs:
 - **Kubelet:** An agent that talks to the Control Plane and makes sure containers are running properly.
 - **Container Runtime:** Software that runs the containers (like Docker or containerd).
 - **Kube-proxy:** Manages network communication inside and outside the cluster.

43) How do you give access the root user on a Linux system via SSH ?

A) add user Mahesh

New password :

vi sudo

Mahesh ALL =(ALL :ALL)NO PASSWD :ALL

cd /etc/ssh/

vi sshd -config

Password Authentication: yes

systemctl restart ssh

44) What is Terraform ?

A) Terraform is a tool that lets you create, change, and manage cloud resources (like servers, networks, databases) using code.

45) what is Difference between git and git hub ?

A)

Git	GitHub
Git is a tool	GitHub is a service/platform
It helps you track code changes locally	It lets you store and share Git repositories online
You can use Git without the internet	GitHub requires an internet connection
Installed on your computer	Accessed via web (github.com)
Maintains code history, branches, versions	Adds collaboration features like pull requests, issues, and team management

46) what is Merge conflicts ,when it comes and how to resolve ?

A) A **merge conflict** happens when **two people change the same part of a file** (like the same line of code), and Git doesn't know which change to keep.

Merge conflicts usually occur when:

- You run `git merge`, `git pull`, or `git rebase`
- And **you and someone else have both edited the same line in the same file**, or deleted/renamed files differently.

How to resolve :

- Open the file with conflict
- Choose the correct version or combine the changes
- Remove the conflict markers (`<<<<<<`, `=====`, `>>>>>>`)
- Then run:
`git add filename`
`git commit`

47) how to create a **private EC2 instance** and give it **private access** ?

A) How to create a private instance:

1. **Launch an EC2 instance** in a **private subnet** (a subnet without an internet gateway).
2. Ensure the subnet has a **route to a NAT Gateway** or **no route to the internet**.
3. Do **not assign a public IP** to the instance.

How to give private access (access the instance privately):

Use a Bastion Host (Jump Server):

- Launch a **public EC2 instance** in a public subnet.
- SSH into the public instance first.

- From there, SSH into the private instance using its private IP.

48) What is Jenkins Version used ?

A) 2.49 or 2.50

49) what is CI / CD ?

A) **Continuous Integration** is a practice where developers integrate or merged code into a shared repository frequently. Jenkins automates this process.

Example : Jenkins builds the project and runs tests whenever code is pushed.

Continuous Delivery means automatically testing and preparing your code so it's always ready to be deployed to production — but the deployment is done **manually** when you choose.

Example : Your code is tested and packaged after every change, and you can deploy it anytime with one click.

Continuous Deployment means every change that passes tests is **automatically deployed** to production — no manual steps.

Example:

You push code → it's tested → it's live on the website without human approval.

50) What are the challenges you faced in project ?

A) I faced an issue with failing Jenkins pipeline due to incorrect environment variables. I analyzed the logs , identified root cause , and updated the pipeline configuration to dynamically pull variables based deployment environments. This improved deployment reliability.

51) Write sample Jenkins pipeline ?

A)

```
pipeline {
    agent any
    stages {
        stage ('checkout') {
            steps {
```

```

        git 'https://github.com/your-username/your-repo.git'
    }
}
stage ('validate') {
    steps {
        sh 'mvn validate'
    }
}
}
}

```

52) What is agent any in pipeline ?

A) An agent is a machine that runs tasks (jobs) as assigned by the master.

53) What is DevOps ? Why is important in IT industry ?

A) **DevOps** is a combination of **Development** and **Operations**. It is a **set of practices** that helps teams **build, test, and release software faster and more reliably**.

Why is DevOps important in the IT industry?

1. **Faster delivery** of software and updates
2. **Improved collaboration** between developers and operations teams.
3. **Automation** of manual tasks (like testing, building, deployment)
4. **More reliable systems** with fewer errors.
5. **Scales easily** with cloud and modern tools.

54) How to check the disk usage in Linux ?

A) df -h

55) why we automates the process in Jenkins ?

A) We automate processes in Jenkins to **save time, reduce errors, and ensure consistency** in building, testing, and deploying applications.

56) How to deploy web application from git to Kubernetes EKS .Explain the process?

A) To deploy a web app from Git to EKS, I use a CI/CD tool like Jenkins. The pipeline pulls the code from Git, builds a Docker image, pushes it to ECR, and then applies Kubernetes YAML files to EKS using `kubectl`. This automates the deployment and makes the app live on the cluster.

57) what is command used to tagging image in docker ?

A) `docker tag docker image id docker hub name /image name:tag`

58) What is Docker compose file ?

A) A **Docker Compose file** is a **YAML file** (`docker-compose.yml`) used to define and run **multiple Docker containers together** as one application.

59) What is Horizontal auto-scaling and Vertical auto-scaling ?

A) **Horizontal Auto-Scaling** : Adds or removes more instances (servers or pods) based on load.

Vertical Auto-Scaling : Increases or decreases the size (CPU/RAM) of an existing instance.

60) If a deployment fails, I follow these basic steps to troubleshoot and resolve the issue ?

A) If a deployment fails, I start by checking the logs and error messages to identify the issue. Then I verify the configuration files (like Docker file, Kubernetes YAML, or Jenkins file), check resource availability, review recent code changes, and ensure that all dependencies and services are running correctly. Once the root cause is found, I fix the issue, test locally or in staging, and redeploy.